Artificial Intelligence (AI) is a branch of computer science focused on building systems that can perform tasks that typically require human intelligence. These tasks include problem-solving, decision-making, language understanding, and perception.

Machine Learning (ML) is a subset of AI that allows systems to learn from data and improve their performance over time without being explicitly programmed. Deep Learning is a further subset of ML that uses neural networks with many layers.

Natural Language Processing (NLP) enables machines to understand and generate human language. Applications include chatbots, translation systems, and sentiment analysis tools.

Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval with text generation. It retrieves relevant documents or data and uses them as context for generating accurate responses. This helps reduce hallucination in large language models.

Large Language Models (LLMs) like GPT are trained on vast amounts of text data and can generate human-like responses. However, they may sometimes produce incorrect or fabricated information if not properly guided.

Vector databases store data as embeddings, which are numerical representations of text. These embeddings allow efficient similarity search, enabling systems to find relevant information quickly.

AI systems must also consider ethical concerns such as bias, fairness, and transparency. Responsible AI practices ensure that systems are safe, reliable, and trustworthy.

In real-world applications, AI is used in healthcare, finance, education, and automation. It helps improve efficiency, accuracy, and decision-making processes.